

Multimodal AI Agent for Suicide Prevention in Sri Lankan Undergraduates

CSCI 43018 –Project Proposal Presentation

M.A.S.Sewwandi

CS/2020/052

Supervisor's name: Ms.Meesha Mervyn

Table of content

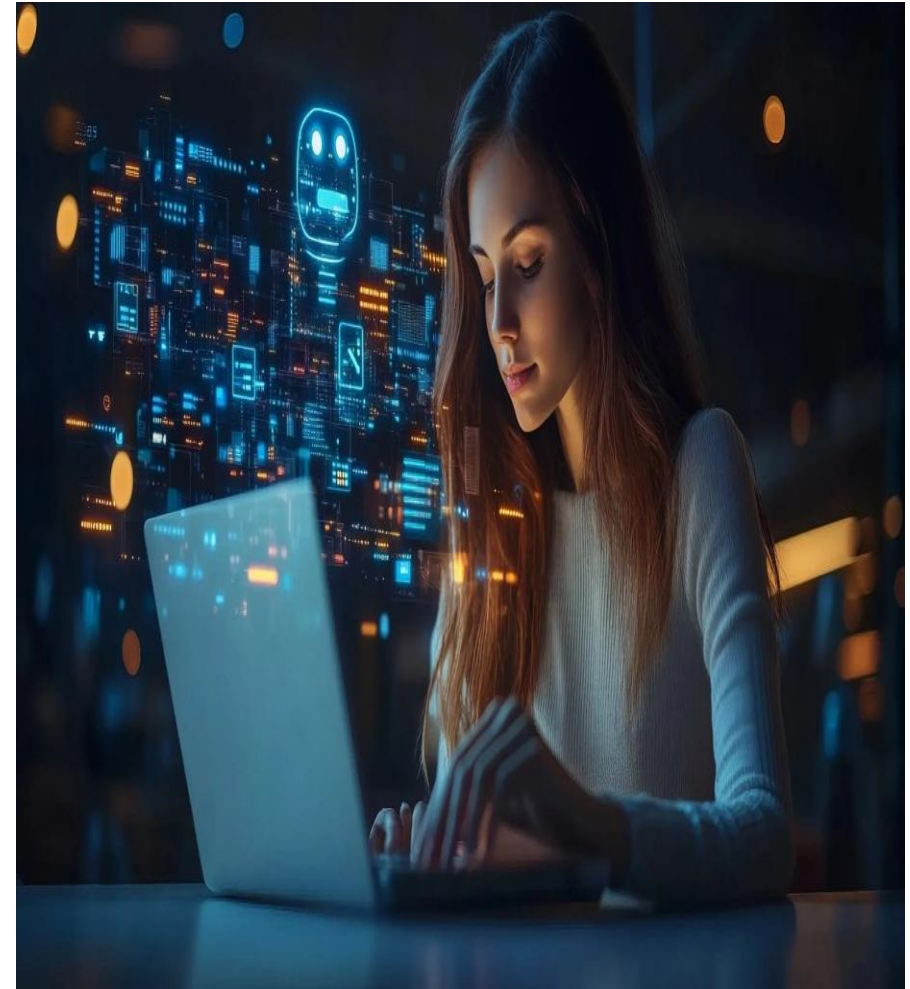
- 01 Introduction
- 02 Research Problem
- 03 Background & Literature Gap
- 04 Aim
- 05 Objectives
- 06 Methodology
- 07 Time Frame
- 08 Ethical Concerns
- 09 References



Introduction

Developing a Multimodal AI Agent for Suicide Prevention in Sri Lankan Undergraduates.

- **Combines text, audio, video, and behavioral signals for continuous monitoring**
- **Provides real-time interventions and supportive resources**
- **Automatically escalates high-risk cases to professional counselors**
- **Works 24/7, private, and culturally adapted for Sri Lankan students**



Research Problem

Suicide is a leading cause of death among young adults worldwide Sri Lankan undergraduate context:

- Academic pressure, financial struggles, competitive job market
- Cultural stigma → students avoid seeking help
- Limited access to professional counseling support on campus

Current system limitations:

- Detection Gap – relies on self-reporting; early signs often missed
- Intervention Gap – support is reactive, not continuous or immediate
- Escalation Gap – high-risk students may remain unnoticed until crisispoint

Core Problem:

Lack of proactive, culturally adapted, and technology-driven suicide prevention tools for Sri Lankan undergraduates



Background & Literature Gap

Current AI suicide prevention research often relies on single-modality data (e.g., text only), creating an incomplete picture and lacking cultural adaptation for Sri Lanka. Furthermore, a systemic review by Bennett-Poynter et al. (2025) highlights a critical gap in both ethical governance and multimodal frameworks for such sensitive applications.

My research directly addresses this by building a culturally-aware, multimodal AI agent that integrates multiple data streams within a robust ethical framework, specifically designed for the Sri Lankan context.

Abstract: As suicide rates increase globally, there is a growing need for effective, data-driven methods in mental health monitoring. This study leverages advanced artificial intelligence (AI), particularly natural language processing (NLP) and machine learning (ML), to identify suicidal ideation from Twitter data. A predictive model was developed to process social media posts in real time, using NLP and sentiment analysis to detect textual and emotional cues associated with distress. The model aims to identify potential suicide risks accurately, while minimizing false positives, offering a practical tool for targeted mental health interventions. The study achieved notable predictive performance, with an accuracy of 85%, precision of 88%, and recall of 83% in detecting potential suicide posts. Advanced preprocessing techniques, including tokenization, stemming, and feature extraction with term frequency-inverse document frequency (TF-IDF) and count vectorization, ensured high-quality data transformation. A random forest classifier was selected for its ability to handle high-dimensional data and effectively capture linguistic and emotional patterns linked to suicidal ideation. The model's reliability was supported by a precision-recall AUC score of 0.93, demonstrating its potential for real-time mental health monitoring and intervention. By identifying behavioral patterns and triggers, such as social isolation and bullying, this framework provides a scalable and efficient solution for mental health support, contributing significantly to suicide prevention strategies worldwide.

Keywords: machine learning; artificial intelligence; suicidal ideation detection; mental health analysis; natural language processing; sentiment analysis; predictive modeling



Academic Editor: Victor C.M. Leung

Received: 7 December 2024

Revised: 31 December 2024

Accepted: 14 January 2025

Published: 20 January 2025

Citation: Allam, H.; Davison, C.; Kalota, F.; Lazaros, E.; Hua, D. AI-Driven Mental Health Surveillance: Identifying Suicidal Ideation Through Machine Learning Techniques. *Big Data Cogn. Comput.* **2025**, *9*, 16. <https://doi.org/10.3390/bdcc9010016>

1. Introduction

1.1. Suicide as a Global Public Health Crisis

Suicide is a pressing global public health issue that significantly impacts individuals across various demographics and regions. The complexity of this phenomenon is shaped by diverse risk factors, cultural contexts, and the effects of globalization. Understanding these elements is critical for developing effective prevention strategies.

Globally, suicide ranks as the third leading cause of death among 15- to 19-year-olds [1].

Background & Literature Gap

Current AI systems effectively detect suicide risk but operate in isolation, lacking automated alert protocols to connect at-risk students with counselors. This creates a critical delay in intervention.

Our Novel Contribution: Closed-Loop System

We introduce an AI-counselor feedback loop that:

- Automatically alerts counselors of high-risk cases
- Generates detailed student mental health reports
- Enables continuous system improvement through counselor input

European Psychiatry

www.cambridge.org/epa

Review/Meta-analysis

Cite this article: Lejeune A, Le Glaz A, Perron P-A, Sebt J, Baca-Garcia E, Walter M, Lemey C, Berrouiguet S (2022). Artificial intelligence and suicide prevention: A systematic review. *European Psychiatry*, 65(1), e10, 1–8
<https://doi.org/10.1192/j.eurpsy.2022.8>

Received: 14 September 2021
Revised: 13 December 2021
Accepted: 20 December 2021

Keywords:
Artificial intelligence; machine learning; neural network; suicide prevention

Author for correspondence:
*Alban Lejeune,
E-mail: alban.lejeune@gmail.com

Present address of Alban Lejeune: Hôpital de Bohars, CHRU de Brest, Brest, France.

Artificial intelligence and suicide prevention: A systematic review

Alban Lejeune^{1*}, Azil Le Glaz¹, Pierre-Antoine Perron¹, Johan Sebt², Enrique Baca-Garcia³, Michel Walter^{1,4}, Christophe Lemey^{1,4,5} and Sofian Berrouiguet^{1,6}

¹URCI Mental Health Department, Brest Medical University Hospital, Brest, France; ²Mental Health Department, French Polynesia Hospital, FPC-11965, Papeete, French Polynesia; ³Departamento de Psiquiatría, IIS Fundación Jiménez Díaz, Madrid, Spain; ⁴EA 7479 SPURBO, Université de Bretagne Occidentale, Brest, France; ⁵SPURBO, IMT Atlantique, Lab-STICC, UMR CNRS 6285, F-29238, Brest, France and ⁶LaTDM, INSERM, UMR 1101, Brest, France

Abstract

Background. Suicide is one of the main preventable causes of death. Artificial intelligence (AI) could improve methods for assessing suicide risk. The objective of this review is to assess the potential of AI in identifying patients who are at risk of attempting suicide.

Methods. A systematic review of the literature was conducted on PubMed, EMBASE, and SCOPUS databases, using relevant keywords.

Results. Thanks to this research, 296 studies were identified. Seventeen studies, published between 2014 and 2020 and matching inclusion criteria, were selected as relevant. Included studies aimed at predicting individual suicide risk or identifying at-risk individuals in a specific population. The AI performance was overall good, although variable across different algorithms and application settings.

Conclusions. AI appears to have a high potential for identifying patients at risk of suicide. The precise use of these algorithms in clinical situations, as well as the ethical issues it raises, remain to be clarified.

Introduction

Overall mortality from suicide is currently about 700,000 deaths per year [1]. Suicide and suicidal behavior are a public health concern. Among people surviving a suicide attempt, about one-third come to the emergency department for help [2]. Suicide risk assessment of these patients is a daily challenge in psychiatric practice. The personal and family history, particularly of suicide attempts, plays a major role in assessing suicide risk. Personal history of suicide attempt is the most significant factor in the risk of death by suicide [2]. Patients with severe mental illness, in particular mood disorder, borderline personality disorder, and anorexia nervosa, are more likely to attempt suicide [3,4]. They are at higher risk of recurrence in the years following the first attempt [5]. Among patients who survived a suicide attempt, a study found that certain subgroups (alcohol consumption, personality disorder, and young age) were also more likely to attempt suicide again, and others were more likely to die by suicide after a first attempt (elderly patients) [6]. In clinical practice, particular attention must also be given to the period of discharge from the care services. The risk appears to be increased within the 2 weeks following discharge from the hospital [2].

Since the incidence of suicide and suicide attempts remains high [1,7,8], we need new approaches to identify and manage patients at high risk of suicide. The current suicide risk assessment methods are based on questioning and therefore subject to subjectivity. Their accuracy and predictive value are limited [9]. Several scales can be used in the suicide risk assessment, but their accuracy seems insufficient [10]. In their meta-analysis, Franklin et al. [11] found that the ability to predict suicide had not improved over the past 50 years. The ability to predict a suicide attempt lack accuracy. Advances in suicide risk assessment are needed [11].

Artificial intelligence (AI) and machine learning (ML) have emerged as ways to increase risk

© The Author(s), 2022. Published by Cambridge University Press on behalf of the European Psychiatric Association. This is an Open Access

Aim

- To design, develop, and evaluate a prototype multimodal AI agent for suicide prevention
- Focus on early detection of suicidal ideation through multimodal analysis
- Deliver real-time, culturally sensitive interventions for undergraduates
- Provide an automated escalation pathway to professional counselors in high-risk situations



Objectives

- **Develop an ethical framework for multimodal data collection with informed consent**
- **Train individual AI models:**
 - NLP for text-based sentiment and despair markers
 - CNN for facial expression recognition
 - Acoustic analysis for vocal stress patterns
 - Behavioral telemetry for online activity changes
- **Implement a late-fusion algorithm to generate composite risk scores**
- **Design a tiered intervention protocol (low, medium, high risk)**
- **Build a secure counselor dashboard for real-time monitoring**
- **Conduct pilot evaluation for technical performance and user acceptance**



Methodology

- **Data Collection:** Pilot study with ~30 undergraduate volunteers
- **Data Modalities:** Text, audio, video, and behavioral signals
- **System Architecture:**
 - Preprocessing layer – cleaning and feature extraction
 - Fusion layer – combining multimodal features
 - Risk detection layer – classifying risk level (low, medium, high)
 - Intervention & escalation layer – delivering responses and alerts
- **Evaluation Strategy:**
 - Quantitative: Accuracy, Precision, Recall
 - Qualitative: Usability testing, student & counselor feedback

Time Frame

Task	Week				
	1-6	7-12	13-20	21-26	27-34
Literature review, gap analysis, research design, and ethics approval					
Dataset collection, preparation, and preprocessing					
AI model training, multimodal fusion, conversational agent development					
System testing, counselor dashboard creation, preliminary evaluation					
Final evaluation, report writing, and dissemination of findings					

Ethical Concerns

- **Informed Consent:** Multi-stage, voluntary opt-in for each modality (text, audio, video)
- **Transparency:** Students can view their data, risk scores, and reasoning behind alerts
- **Handling False Positives:** Responses designed to be supportive, not alarming or punitive
- **Minimizing Harm:** Protocols developed with counselors to ensure safe, ethical handling



References

- [1] World Health Organization, Suicide worldwide in 2021: global health estimates. Geneva: World Health Organization, 2025. doi: 978-92-4-011006-9 (eISBN). Licence: CC BY-NC-SA 3.0 IGO. [Online].
- [2] T. Rajapakse, “Suicide in Sri Lanka: Reflections and ways forward,” Sri Lanka Journal of Psychiatry, vol. 15, no. 2, pp. 1–5, Dec. 2024, doi: 10.4038/sljpsyc.v15i2.8602.
- [3] S. D. Amarasuriya, N. J. Reavley, A. Rossetto, and A. F. Jorm, “Helping intentions of undergraduates towards their depressed peers: a cross-sectional study in Sri Lanka,” BMC Psychiatry, vol. 17, no. 1, p. 40, Dec. 2017, doi: 10.1186/s12888-017-1192-7.
- [4] H. Allam, C. Davison, F. Kalota, E. Lazaros, and D. Hua, "AI-Driven Mental Health Surveillance: Identifying Suicidal Ideation Through Machine Learning Techniques," Big Data and Cognitive Computing, vol. 9, no. 1, p. 16, Jan. 20, 2025.
- [5] L. Bennett-Poynter, S. Kundurthi, R. Besa, D. W. Joyce, A. Kormilitzin, et al., “Harnessing digital health data for suicide prevention and care: A rapid review,” Digital Health, vol. 11, p. 20552076241308615, Jan. 2025. 8
- [6] A. Lejeune, A. Le Glaz, P.-A. Perron, J. Sebti, E. Baca-Garcia, and others, "Artificial intelligence and suicide prevention: A systematic review," European Psychiatry, vol. 65, no. 1, p. e19, 2022.
- [7] P. Bufano, M. Laurino, S. Said, A. Tognetti, and D. Menicucci, "Digital phenotyping for monitoring mental disorders: Systematic review," Journal of Medical Internet Research, vol. 25, p. e46778, Dec. 13, 2023, doi: 10.2196/46778.



Thank You